

Parth Mital

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Education

Vellore Institute of Technology (VIT)
B.Tech in Computer Science & Engineering
CGPA: 8.70 / 10

Vellore, India
2023 – 2027

Technical Skills

Languages: Python, Java, C, C++, TypeScript, JavaScript

Web & Full-Stack: React.js, Next.js, Tailwind CSS, SQL, PostgreSQL, MongoDB, REST APIs

Artificial Intelligence: Machine Learning, Deep Learning, Natural Language Processing

Automation & Scripting: Web Scraping, Task Automation

Projects

NeuroScan AI - Brain Tumour Analysis (Full-Stack AI & Medical Imaging Platform)

🌐 github.com/parthmital/NeuroScan-AI

- Developed an end-to-end system with JWT-based authentication, secure NIfTI upload & automated pipeline (detection → classification → segmentation) along with dashboard & 3D overlay visualisation.
- Trained 2× ResNet50 models on 46,920 images & 1× 3D U-Net (BraTS2020 dataset); achieved 80.07% detection accuracy (85.99% AUC), 87.41% classification accuracy, Dice scores: WT 0.847, TC 0.753, ET 0.761.
- Tech: Python, FastAPI, TensorFlow, PyTorch, OpenCV, NiBabel, scikit-image, SQLAlchemy (SQLAlchemy + Pydantic), SQLite, JWT, bcrypt, React 18, TypeScript, Vite, Tailwind CSS, Recharts.

VIT Carpool - Campus Ride-Sharing Platform (Full-Stack Web Application)

🌐 vitcarpool.vercel.app 🌐 github.com/parthmital/VIT-Carpool

- Replaced unstructured WhatsApp carpool groups with a structured, searchable platform.
- Secure Google OAuth authentication, user profiles & ride creation with source, destination, date, time.
- Real-time seat tracking, advanced search/filtering, full CRUD for rides & participants, responsive design.
- Tech: React, TypeScript, Vite, Tailwind CSS, Supabase (PostgreSQL, RLS), REST API.

VIT Resource Automation Pipeline (Automation & Web Scraping)

🌐 github.com/parthmital/VIT-Resource-Automation-Pipeline

- Replaced weeks of manual VTOP & UniBud resource collection with a fully automated download pipeline.
- Enabled objective faculty comparison using page & slide counts & generated a uniform PDF-only repository.
- Tech: Python, Selenium, Playwright, pypdf, pywin32, comtypes.

OS Debloating Performance & Privacy Study (Systems Performance Analysis)

🌐 github.com/parthmital/Operating-System-Debloating

- Conducted a VM-based study on multiple Windows & Linux OS builds to analyse the impact of debloating.
- Measured boot time, idle RAM usage, CPU utilisation, disk I/O, & outbound network connections across 30+ runs per OS.
- Reduced idle RAM by 15-60%, boot time by 10-48%, & blocked 50-100% of unnecessary telemetry endpoints.

Experience

Web Development Intern - Hiraeth Tech

Jun 2025 – Jul 2025

Vadodara, India

- Developed 5+ responsive WordPress websites using Elementor & custom CSS.
- Optimised 40+ media assets, reducing page load time by 30%.